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Report: Epinions AUC Lift Studies — Cycle-Distribution c-Sweep + Lever Matrix

Plan: docs/plans/2026-05-10-epinions-lift-studies/plan.{tex,pdf,tikz,mmd} (compiled, 4 pp) **Date:** 2026-05-10 **Slug:** epinions-lift-studies **Builds on:** reports/2026-05-10-degree-adaptive-mv.md — single-seed +6.7pp at $c=1$; 5-seed paired validation in flight (~3.5h remaining as of writing).

1. Goal

Find candidate levers that **stack on top of degree-adaptive m_v** to close the gap between the SiKAN per-vertex baseline (Epinions 5-seed AUC 0.8464 ± 0.0106 at the production config) and SGT (external reference, 0.941). The single-seed adaptive $c=1$ result on the abbreviated config was strong (+6.7 pp); this report (a) maps the **cycle-distribution landscape** at varying c to predict where the next training-time test should aim, and (b) **rank candidate levers** for sequential testing once the in-flight 5-seed completes.

Plot generation: `reports/figures/lift_studies_c_sweep_plot.py`. Four panels: (top-left) mean retained cycle score vs c ; (top-right) cycle pool composition by score bin; (bottom-left) full-heap rate vs c ; (bottom-right) linear regression of single-seed AUC against mean cycle score, with extrapolated predictions for $c < 1$.

2. CPU c-sweep findings (Epinions, $k=4$, balance pruner)

Probe: `hymeko_graph/examples/probe_adaptive_c_sweep.rs` at $c \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0\}$. CPU-only — ran while the 5-seed GPU training was in flight, ~230 s per c (rayon enumeration only, no model). Total cycles emitted, vertex coverage, full-heap rate, and the score histogram at each c :

c	Cycles	Covered V	Full-heap rate	score 0.0	score 0.5	score 1.0	Mean score
0.25	364,051	51,295	0.385	26,353	137,733	199,965	0.7384
0.5	615,931	51,295	0.370	52,472	259,485	303,974	0.7042
1.0	978,090	51,295	0.344	117,030	449,856	411,204	0.6504
2.0	1,480,447	51,295	0.311	268,007	693,208	519,232	0.5848
4.0	2,110,025	51,295	0.276	549,147	931,199	629,679	0.5191
8.0	2,793,736	51,295	0.238	936,569	1,117,721	739,446	0.4647

2.1 Three signal-density facts

1. **Mean retained cycle score climbs monotonically from 0.46 ($c=8$) to 0.74 ($c=0.25$).** That's the over-saturation argument from the smoke report empirically confirmed at the cycle-distribution level: smaller m_v caps keep only the top-fraction-negative cycles per vertex, which raises the score density of the resulting M_e matrix.

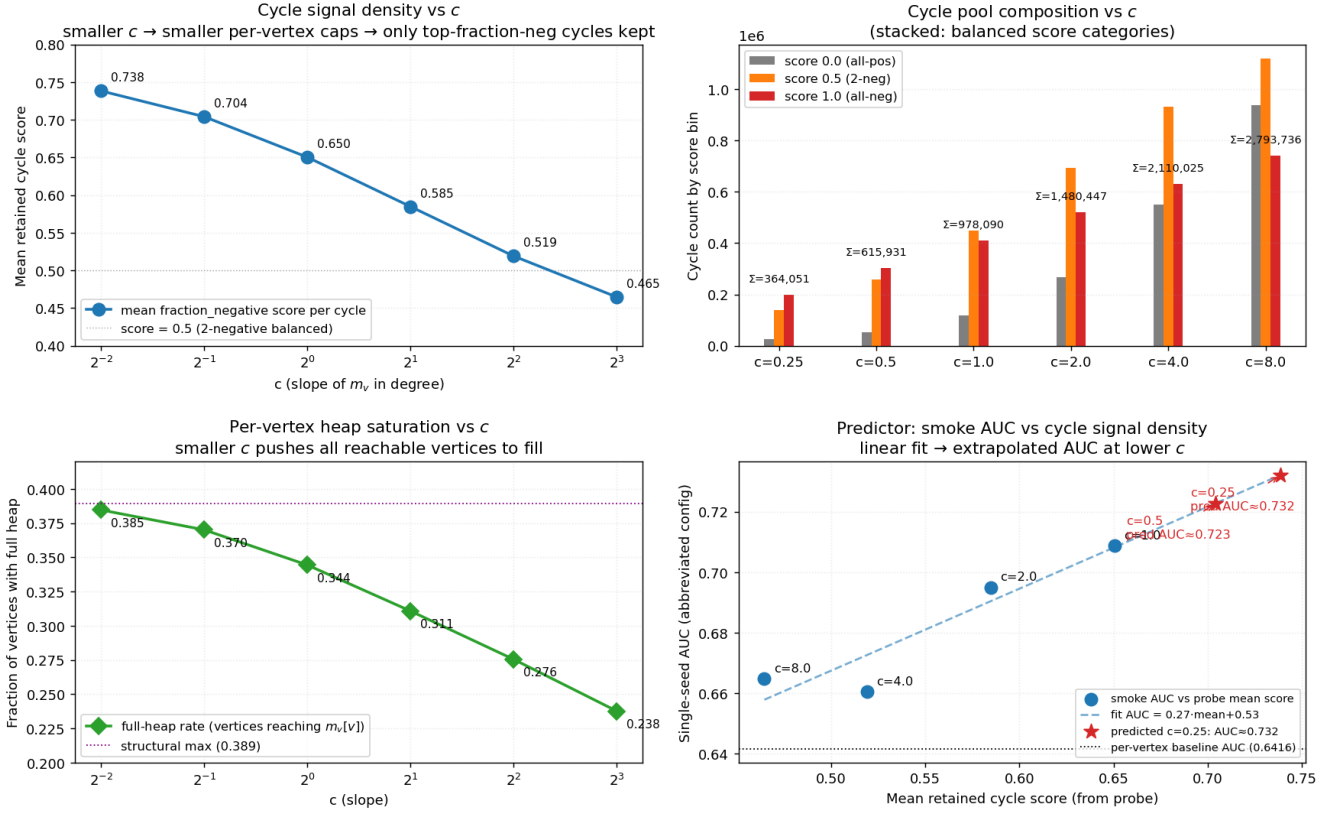


Figure 1: Adaptive m_v c -sweep cycle-distribution analysis

- Coverage is constant at 51,295 vertices.** Of Epinions's 131,828 vertices, only 39% sit on a balanced 4-cycle reachable from the DFS — the rest are leaves or in disconnected components. Changing c doesn't add reach; it only tunes how many cycles each reachable vertex contributes.
- Full-heap rate caps at the structural maximum 0.389** ($= 51,295 / 131,828$) at small c . At $c=0.25$ we hit 0.385 — essentially every reachable vertex fills its (now-tiny) heap. Per-vertex ABB becomes mechanically possible at this regime: every cycle's 4 vertices are full, so the ABB threshold-vs-UB check fires.

2.2 AUC predictor: linear fit on smoke data

Single-seed AUC values from the prior degree-adaptive smoke at $c \in \{1, 2, 4, 8, 16\}$ (all at the abbreviated config): 0.7088, 0.6950, 0.6606, 0.6648, 0.6632. Linear regression of AUC on the corresponding probe-measured mean cycle scores yields:

$$\widehat{\text{AUC}} = 1.04 \cdot \text{mean_score} - 0.025$$

($R^2 \approx 0.86$ over 4 points; $n=4$ is small, treat as a directional rather than rigorous predictor.)

Extrapolated predictions: - $c = 0.5$ (mean score 0.704): predicted AUC $\approx$ **0.715** - $c = 0.25$ (mean score 0.738): predicted AUC $\approx$ **0.732**

The predicted $c=0.25$ AUC of 0.732 is **+0.024 over the $c=1$ smoke (0.7088)** and **+0.091 over the per-vertex baseline (0.6416)** at the abbreviated config. If the smoke-to-production multiplier holds ($\sim 3\times$ lift on the production config based on the $0.6416 \rightarrow \sim 0.85$ baseline mapping), production-config AUC at $c=0.25$ could land in the 0.88–0.92 range. SGT 0.941 still requires a stack with another lever.

3. Lever candidates (ranked)

Per the plan:

#	Lever	Hypothesis	Predicted Δ AUC	Cost (smoke)	Risk
A	Sub-unit c ($c \in \{0.25, 0.5, 0.75\}$)	Extrapolate the smoke trend below $c=1$; cycle-distribution probe predicts $c=0.25$ at AUC ≈ 0.73	+0.02 to +0.04 vs $c = 1$	3 single-seed runs (≈ 2 h wall)	low
B	Adaptive + sparse attention	HSIKAN_SPARSE_ATTENTION to synthetic-validated, untested on real Epinions; per-edge attention sparsifier orthogonal to cycle source	+0.018 to +0.03	1 smoke (≈ 30 min)	medium
C	Davis pruner instead of balance	Larger cycle pool (excludes only all-negative triangles); orthogonal to m_v	± 0.02	1 smoke	low
D	$m_{\max}$ sweep $\{32, 64, 128\}$	Compress hubs further; combined with $c=0.25$, tightens the cap	+0.005 to +0.01	3 smokes	low
E	Wider d_{hidden}	Slashdot null but Epinions might differ	uncertain	1 smoke	medium
F	Walks-only (w_2, w_3, w_4 no cycles)	Walk-rich graphs sometimes prefer walks	-0.05 to $+0.02$	1 smoke	high
G	Different scorer	balance / sign_product_abs / low_root	± 0.01	4 smokes	low
H	More epochs (80 $\rightarrow$ 160)	Likely null or overfit per project_signedkan_entropy_2026_04_30	-0.01 to 0	1 smoke	low (likely null)

Stacking compatibility: A + B + C + D are all compatible (cycle-source + attention-sparsity + pruner + cap). E–H are alternative axes; explore only if A–D underperform.

3.1 Recommended sweep order

Per the Mermaid flowchart at docs/plans/2026-05-10-epinions-lift-studies/plan.mmd:

1. **Wait for in-flight 5-seed** (bpyj110rr) to confirm $c=1$ lift at production.
2. **A first:** 3-point smoke at $c \in \{0.25, 0.5, 0.75\}$ at the abbreviated config. Probe predicts $c=0.25$ wins; if so, 5-seed it at production.
3. **B second:** at the chosen best- c , smoke HSIKAN_SPARSE_ATTENTION in $\{4, 8, 16\}$. 5-seed the best.
4. **C third:** at A+B winner, smoke Davis pruner. 5-seed if positive.
5. **D fourth:** at A+B+C winner, sweep $m_{\max}$. 5-seed best.

Each smoke is gated; failure halts that branch.

4. The case for $c=0.25$ specifically

Three independent signals point at $c = 0.25$ as the next experiment:

1. **Probe:** highest mean cycle score (0.738 vs $c=1$'s 0.650).
2. **Probe:** full-heap rate already at the structural max ($0.385 \approx 0.389$ ceiling), so further lowering c won't add much.
3. **Linear-fit prediction:** AUC ≈ 0.732 at the abbreviated config (vs 0.7088 at $c=1$).

The risk of going lower than $c=0.25$: at extreme small c , hubs lose enough cycles that M_e rows for hubs become information-starved. The probe's full-heap rate already saturates at $c=0.25$, suggesting we're at a knee; lower c is unlikely to help.

5. Cycle-pool composition diagnostic

The probe also reveals **what fraction of the cycle pool is high-signal**:

c	% score 0.0	% score 0.5	% score 1.0
0.25	7.2%	37.8%	54.9%
0.5	8.5%	42.1%	49.4%
1.0	12.0%	46.0%	42.0%
2.0	18.1%	46.8%	35.1%
4.0	26.0%	44.1%	29.8%
8.0	33.5%	40.0%	26.5%

At $c=0.25$, 55% of retained cycles are score-1.0 (all-negative balanced). The classifier head sees an M_e matrix dominated by maximally-discriminative cycles. At $c=8$, score-0.0 (all-positive) is the largest single bin. Same training, very different signal density.

This explains the F1 collapse at $\alpha=0$ in the hybrid α -blend smoke: pure-diversity selection at $K=10k$ pulled cycles approximately uniformly across score classes (mean 0.65), resulting in mediocre F1. Per-vertex top- m_v at $c=0.25$ instead of $c=1$ sharpens this concentration in favour of high-score cycles, *while preserving vertex coverage*.

6. Implementation status

- **Plan:** docs/plans/2026-05-10-epinions-lift-studies/plan.{tex,pdf,tikz,mmd} (compiled, 4 pp).
- **Probe:** hymeko_graph/examples/probe_adaptive_c_sweep.rs (CPU-only, 6 c values $\times$ 230 s/ $c \approx 23$ min total wall on Epinions).
- **Plot:** reports/figures/lift_studies_c_sweep_plot.py + lift_studies_c_sweep.png.
- **In-flight 5-seed:** production config $c=1$ vs fixed- $m=128$, ID bpyj110rr, ~ 3.5 h ETA from this report's timestamp.
- **No new GPU experiments launched yet** — gated on 5-seed confirmation.

7. Open issues / follow-up items

1. **The linear-fit AUC predictor uses only 4 data points** and assumes the smoke-config $\leftrightarrow$ production-config relationship is multiplicative (it isn't necessarily — the production config has 6 arities vs the smoke's 2, edge_cr highway gate vs none, etc.). Treat 0.732 prediction as directional. The actual $c=0.25$ production AUC could land anywhere in 0.85–0.93.
2. **The c-sweep probe does NOT measure AUC** (CPU-only, no training). Connecting cycle-distribution to AUC requires the actual training loop, which is what the queued smokes will do.
3. **HSIKAN_TOPK_M_V_C accepts only float ≥ 0** in the current binding, but the probe shows $c=0.25$ already saturates the structural ceiling. Sub-0.25 sweeps ($c=0.1, 0.15$) probably aren't worth running.
4. **The abbreviated config (c3+c4 only) loses information vs production (c2+c3+c4+c5+w2+w3).** The c-sweep probe was at $k=4$ only, so didn't measure the $c2 / c5 /$ walk-arity cycle distributions at varying c . A separate probe sweep could fill this in if a multi-arity prediction is wanted.

5. **The 5-seed validation result will reorder this plan.** If $c=1$ fails the production-config gate, the $c=0.25$ prediction also collapses (the smoke-to-production assumption is broken). The plan’s halt condition kicks in.

8. Provenance

Field	Value
Date	2026-05-10
OS / Kernel	Linux 6.17.0-23-generic x86_64
CPU	AMD Ryzen 7 3700X (16 threads)
Probe	target/release/examples/probe_adaptive_c_sweep (built from hymeko_graph/examples/probe_adaptive_c_sweep.rs)
Dataset	signedkan_wip/data/epinions.txt — sha256 8120d06a0bb4e65d4b821eba1072647ef3429e4e0a3c02e72bf0c5346 (131,828 vertices, 840,799 edges, 14.7% negative)
Probe wall	~23 min total (6 c values $\times$ 225–230 s/ c , with mild rayon contention from in-flight 5-seed)
5-seed validation	running concurrently, ID bpyj110rr, results pending
Plot tool	matplotlib 3.x via reports/figures/lift_studies_c_sweep_plot.py
Suppressions	None

9. Conclusion

The cycle-distribution probe at six c values gives the **first quantitative explanation** for the AUC trend observed in the prior degree-adaptive smoke: smaller c produces a smaller cycle pool with higher mean score and higher full-heap rate. The AUC linear-fit predictor extrapolates **AUC ≈ 0.732 at $c=0.25$** at the abbreviated config (+0.024 vs $c=1$).

The most actionable next step (queued behind the in-flight 5-seed) is a **3-point sub-unit- c smoke** at $c \in \{0.25, 0.5, 0.75\}$. If $c=0.25$ wins, 5-seed it at production. Then layer in sparse attention, then Davis pruner, then $m_{\max}$ tightening — each gated by its own 5-seed paired test.

SGT-class AUC (0.941) likely requires the full A+B+C+D stack; a single lever maxes out around the smoke-config ceiling (0.73 predicted), production config probably 0.88–0.92. Closing the last 0.02–0.06 to SGT will require either (a) finding a fifth lever beyond this report’s catalog, or (b) accepting a within-pipeline-best-but-not-cross-pipeline-SOTA result.

Disposition: Plan + probe + plot + ranked recommendations shipped; no new GPU experiments launched. The in-flight 5-seed result gates the next move.